Operations architecture decision log

Updated 21 October 2026. Accepted means the product/architecture direction was explicitly established in this project conversation; it does not mean the capability is implemented, audited or installed. Each accepted ADR is versioned in Git. A changed decision gets a new ADR that names what it supersedes. Implementation evidence stays in IMPLEMENTATION-STATUS.md and the milestone gates.

Current runtime: the operator resumed Docker and development on 21 October. The native application runs with explicit local node budgets and current-user grants. The overnight heartbeat remains paused. ADR0017 retains the historical restriction; RESOURCE-REVIEW.md records the supervised resumption. Word/PDF exports are refreshed; see EXPORT-STATUS.md. Full unattended pipeline acceptance remains open.

ADR	Decision	Status	Delivery gate
0001	An OSHAL experience slice inside the existing Intelligent Operations package; one OSHAL login, standard connections and canonical tickets.	Accepted design	M1 native experience and local current-caller acceptance passed; background source activation remains M2.
0002	Expose raw deliveries, separate OTLP observations and durable explanations of filtering, delivery deduplication, symptom reduction and correlation.	Accepted design	M2/M4; established algorithm and stored-query/window backend acceptance.
0003	Synchronize linked ServiceNow, GitHub, GitLab and Jira records through the standard connector framework with explicit field ownership and guarded writes.	Accepted design	M5b; shared operation/webhook gaps must be completed before live activation.
0004	Reuse user-bound OSHAL connections for CLI tools, discovery and evidence; explicit scoped activation for unattended work.	Accepted design	M2/M3/M7; wrong identity, expiry, revocation and no-default-fallback acceptance.
0005	Deploy a persistent native OSHAL experience in the isolated Kubernetes lab; authenticated manual captures precede continuous source activation.	Implemented first increment	Native login/roles, captures and persistence; fresh kernel-first/public-only installation and desktop portal login verified. Codex/Claude transfer subsequently verified; see ADR0006.

ADR	Decision	Status	Delivery gate
0006	Select independent Operations, Office and email applications with built-in Jarvis; configure Telegram on the shared notification rail.	Partially delivered	Package/access and real account/Office acceptance; mailbox sign-in and actual external alert delivery remain separate gates.
0007	Durable reduction before enrichment; a separate anomaly detector feeds the same decision pipeline.	Accepted design	Source review complete for the named stages; distributed services and detector acceptance remain open.
0008	Reuse the reviewed host/service, cloud network and Kubernetes topology structure with scoped native identities and completed scan generations.	Accepted design	M3/P4; source trace in GRAPH-STRUCTURE.md, bounded native temporal graph and replay correlation accepted; scheduled inventory remains open.

ADR records

- adr/0001-oshal-experience-slice.md
- adr/0002-explainable-event-decisions.md
- adr/0003-standard-ticket-synchronization.md
- adr/0004-user-bound-tool-connections.md
- adr/0005-native-kubernetes-first-increment.md
- adr/0006-hackathon-communications-baseline.md
- adr/0007-durable-pipeline-and-anomaly-boundaries.md
- adr/0008-source-derived-temporal-topology.md

Open decisions and release constraints

Item	State and next action
Hackathon date and provider accounts	Three-week improvement horizon supplied; exact submission date and provider accounts remain unconfirmed. Priorities are in THREE-WEEK-PLAN.md.
First live ticket-sync provider	Select after native connector capability and disposable-account acceptance; all four remain target coverage.
Production-grade source identity and app store	Native person-scoped store and imports verified; transactional delivery ledger accepted; background source authentication remains required.
Exact native approval/runner coverage	Verify admitted artifact, target, principal and expiry binding before execution.
Demo remediation action	Still requires an explicitly scoped registered action; current demo bundle only checks readiness.
Provider operation gaps	Shared-framework work, not private app adapters. Source declarations alone do not prove runtime availability.
Release/publication	Source published through normal gate: 83 pass, 0 fail, 6 partial and 12 explicit prerequisite skips. First-increment installation verified; package audit and full kernel certification remain pending.

Review and change process

Record the decision, reason, alternatives rejected, consequences, owner boundary and acceptance in an ADR before implementing a changed architectural boundary. Link work to the ADR and milestone. Update plan, proposal, Visio, exported documents and traceability when the intended experience changes. Keep simulated UX acceptance distinct from native and live-provider acceptance. Never mark a decision implemented merely because its mockup or document was created.

• [ADR0009 — Native temporal graph journal](adr/0009-native-temporal-graph-journal.md): implement the reviewed structure on caller-scoped SQL with immutable scans, bounded reconstruction and recorded correlation evidence.

• [ADR0010 — Immutable percolation and reservation](adr/0010-immutable-percolation-and-reservation.md): actual stored-query matching precedes transactional symptom counts and evidence-work reservation through the existing native engine relay.

• [ADR0011 — Standard workspace and workflow artifacts](adr/0011-standard-workspace-and-workflow-artifacts.md): restore the existing shared PVC installation contract; workflow-generated files remain distinct from supplied script inputs and report approval from execution.

• [ADR 0012 — Application-defined cockpit navigation](adr/0012-admitted-application-toolbar.md)

• [ADR 0013 — Inspectable graph and processing cohorts](adr/0013-inspectable-processing-and-graph.md)

• [ADR0014 — Native background processing](adr/0014-native-background-processing.md)

• [ADR0015 — Baseline workspace editor](adr/0015-baseline-workspace-editor.md)

• [ADR0016 — Product and foundation boundary](adr/0016-product-and-foundation-boundary.md): TriageNorth is the submission; inherited platform services remain attributed dependencies.

• [ADR0017 — Local runtime resource gate](adr/0017-local-runtime-resource-gate.md): record the original runtime stop and host/node/workspace budget boundary. Kernel capacity fixtures pass; supervised startup now passes the local placement/reserve checks. A complete production resource profile remains open.

• [ADR0018 — Replay evaluation and bounded local analysis](adr/0018-replay-evaluation-and-local-analysis.md): freeze dataset/rule provenance and independent evaluation, compare statistical baselines and small local models, and retain deterministic source identity and approval boundaries. Dataset preparation, the BGL percolation baseline and native Scenario lab are complete; NAB anomaly/model evaluation remains open.

• [ADR0019 — Observed Systems experience](adr/0019-observed-systems-experience.md): reuse the current user's inventory graph and event page for Kubernetes and other system inspection. Exact identities, historical health and partial observations remain visible. Source, twelve focused checks and package deployment/publication are complete; separate Systems-specific live two-user browser acceptance remains open.

• [ADR0020 — Retained context](adr/0020-three-bank-context-and-standard-streams.md): three logical context banks, minimum-age rotation, safe growth, standard stream mappings and durable preprocessed packets. Native replay, actual 909.788-second retention/packet survival and temporary standard Collector export are accepted. Saved receipts reopen without rerunning events; continuous delegated producers remain open.

• [ADR0021 — Percolation rule workbench](adr/0021-percolation-rule-workbench.md): edit rules/window/capture choices, trial saved candidates without changing active incident state, and explicitly activate against the loaded source revision. Source-focused SQL/OpenSearch/browser checks pass; native rollout follows.

0022	Bounded 3D snapshot explorer and predefined operational graph queries; existing current-user topology authority.	Implemented/browser tested; native acceptance pending	[ADR0022](adr/0022-bounded-3d-snapshot-explorer.md).
0023	Bind preprocessed evidence reservations to canonical original-owner workflows and count actual native outputs separately from recovery.	Source and isolated browser/SQL tested; native pending	[ADR0023](adr/0023-preprocessed-canonical-investigations.md).

• [ADR0024](adr/0024-shared-service-standing.md): show native service selection, personal invocation standing and registered notification routes separately; never infer delivery from credentials or configuration.

• [ADR0025](adr/0025-manual-workflow-ticket-handoff.md): immutable own workflow links, reviewed confirmed pushes, durable unknown-outcome holds and explicitly operator-reported readback.

• Communications acceptance reuses the shared native notifier (core ADR188), with separate original-person SMS/voice addresses and an explicit default-off brief-call flag. A durable operator acceptance ledger reserves before sending, holds unknown outcomes and prevents a repeated test from the same approval. Provider acceptance and human receipt remain separate claims; the one-SMS/one-brief-call/one-Telegram approval does not enable general automatic notifications.

• [ADR0026](adr/0026-local-service-acceptance.md): record three one-time transport tests, muted routes, explicit legacy-response reconciliation, source-pinned local development deployment and the distinction from outstanding full kernel certification/live Jira validation.

• [ADR0027 — Visible decision chain](adr/0027-visible-decision-chain.md): exact admitted inputs, named bot handoffs, graphical ticket progress, reference cost and Create/mail entry journeys; installed UI acceptance passed, fresh monitoring investigation and linked deadline retry tracked separately.

• [ADR0028](adr/0028-inline-bot-permissions.md): declare the Operations conversational bot bindings under existing own-scope read permission; verify Jarvis separately from HTTP and native queued-work acceptance.

• [ADR0029](adr/0029-create-suite-and-native-report-artifacts.md): install Create's declared Video, Publishing and PDF components; retain exact private scratch permissions and distinguish actual native exports/provider acceptance from an automatic report chain.

ADR0030 — Git-backed Jarvis promotion (proposed)

[ADR0030](adr/0030-jarvis-delegation-and-git-promotion.md) and [H1–H5 roadmap](SWARM-PROMOTION-ROADMAP.md) define lower-environment validation and separately authorized upper-environment promotion. This is roadmap scope, not installed functionality.

ADR0031 — Response paths and sustained verification (proposed)

[ADR0031](adr/0031-response-paths-and-verification-branch.md) separates fast mitigation from permanent repair, preserves a Git-backed expedited path and defines a bounded workflow verification branch. Four-screen mockups use only synthetic data.

ADR0032 — Operator resource workbench

[ADR0032](adr/0032-operator-resource-workbench.md) replaces repeated Systems cards with an exact-identity tree/detail view and gives the native cockpit sole navigation ownership when embedded. Source checks pass; installation acceptance is tracked separately.

ADR0033 — Historical RCA references

[ADR0033](adr/0033-canonical-rca-reference-library.md) adds a current-user read projection of canonical reports and their source citations. It does not infer recovery or establish an automatic case-correlation index.

• [ADR0034: Selected pod command console](adr/0034-selected-pod-command-console.md) — shared original-user transport; application review and bounded evidence analysis.

• [ADR0035: Scheduled retained evidence](adr/0035-scheduled-retained-evidence.md) — explicit capture opt-in, atomic packet completion and bounded retries on the existing personal schedule.

• [ADR0036: Operator screen hierarchy](adr/0036-operator-screen-hierarchy.md) — consistent Overview, Systems, pipeline views and source selection, using existing native navigation and authority.

• [ADR0037: Approved response and verified recovery](adr/0037-approved-response-and-verified-recovery.md) — byte-exact generated files, approval bound to digests/target/revision with expiry, one reserved dispatch per approval through the person's own confirmed pod exec, measured recovery outcomes and a server-read ticket update.

• [ADR0038: Local anomaly service, automatic correlation and tick notifications](adr/0038-local-anomaly-service-and-tick-automation.md) — two deterministic detectors with explicit readiness and durable cursors, candidates through the normal intake, opt-in automatic correlation on the tick, notification requests; NAB pilot measured, not tuned.

• [ADR0039: Node-level explorer and coached commands](adr/0039-node-explorer-and-coached-commands.md) — node capacity, pressure, versions, taints and placed pods in Systems; coached kubectl/cloud/in-pod commands reviewed and run under the operator's own authority, results recorded with provenance. KERNEL-RAILS-NEEDED.md lists what the kernel must add for in-application cloud execution, git promotion and swarm-to-swarm.

• [ADR0040: Git promotion and break-glass reconciliation](adr/0040-git-promotion-and-break-glass-reconciliation.md) — the plan's exact files and a manifest go to a new branch with a pull/merge request and optional pipeline; an emergency execution must be reconciled in Git within 24 hours; one reservation per plan, path and destination; provider writes through the person's own confirmed connection (kernel catalog 0f86a3c).

ADR 0001: Operations is an OSHAL experience slice

Status: Accepted design, 20 October 2026. Implementation and installed acceptance remain tracked separately.

Context: The application should feel complete and focused while reusing OSHAL login, granular grants, connections, tickets, tools, approvals and audit. A separate authentication or connection framework would fragment user authority and duplicate shared behavior.

Decision: Evolve the existing intelligent-operations package into an independently navigable OSHAL experience slice. Preserve its specialist identities and canonical ticket workflow. Render Operations views of shared records and current-caller connections; do not create an alternate login, secret store, privileged connector runtime or ticket engine. Shell/navigation ownership does not change authority.

Alternatives rejected: a new product identity/login; a second app duplicating the existing incident bots; application routes that bypass shared connector write confirmation or tool authority.

Consequences: Native experience/current-caller APIs must be verified. Shared gaps belong at their owner boundary. The app owns sources/mappings, operational inventory, decisions, evidence and the domain UX. No manifest permission or connector allow-list is expanded just because a prototype displays a provider.

Acceptance: Signed-out access refuses; the signed-in caller sees only admitted records/connections; another user cannot inherit the first user's grants; revocation clears access; existing bot/ticket IDs stay intact. A browser role selector is never accepted as server authorization.

ADR 0002: Every event decision is inspectable

Status: Accepted design, 20 October 2026. Production reduction and storage remain planned.

Context: Operators need to see why incoming data was filtered, quarantined, grouped or dispatched. Alert deliveries and telemetry are different records, and repeated deliveries differ from independent occurrences. An opaque incident total cannot demonstrate the established noise-reduction behavior.

Decision: Keep immutable raw-delivery provenance, separate OTLP observations, normalized alerts and versioned processing decisions. The data view traces source, mapping/rule versions, scoped identity, filter reason, delivery retry, symptom tuple, suppression/reservation, correlation membership and conditional snapshot outcome. Replays link old and new interpretations without overwriting history.

Preserve the established production algorithm: stable scoped keys, suppression windows, reserve-before-dispatch, release-on-failure and bounded correlation. Inspect the existing implementation before integrating it. The prototype's simple severity/tuple demonstration is not that implementation. Stored-query percolation and telemetry window evaluation remain separate backend capabilities.

Alternatives rejected: deduplication by title/hostname/message; treating every measurement as an alert; filtering after costly investigation; conflating cross-source correlation with delivery deduplication; claiming a local fixture demonstrates durable or concurrent correctness.

Consequences: Decisions need durable scope, input references, versions, outcome/reason, incident link, counts, time and replay lineage. Missing identities stay quarantined; unauthorized content is withheld. A conflicting payload reusing a delivery ID must not be silently accepted as a retry.

Acceptance: Explain duplicate ID versus repeated occurrence; preserve distinct source/type/resource keys; test 100 occurrences, concurrent reservation, crash/restart and release-on-failure; prove no costly analysis before reduction; inspect unmatched/quarantined records and separate OTLP signals.

ADR 0003: External ticket synchronization uses standard connectors

Status: Accepted design,20 October2026. ADR0025 implements a manual outbound handoff in source; unattended bidirectional synchronization remains open.

Context: Inbound monitoring integrations and outbound ITSM/developer-ticket integrations serve different purposes. Teams need linked ServiceNow incidents, GitHub/GitLab issues and Jira issues while retaining one canonical OSHAL ticket. All use the signed-in user's standard connection authority.

Decision: Configure inbound alert sources separately from outbound and optional bidirectional ticket sync. Reuse the standard connector catalog, per-user/named-account selection, shared-connection grants, secret references, confirmed write boundary, webhook verification and audit. Implement missing operations in that framework rather than direct app HTTP clients or a second credential system.

A durable link identifies tenant/principal scope, canonical ticket, provider, connection/account, external record and revisions. Policy owns field direction, status mapping, comment handling and conflict resolution. Track origin markers and revisions to suppress echoes. A durable operation ledger reserves stable operation identity before writes. Reconcile unknown creates before retrying; if a provider cannot prove the outcome, retain a blocked/manual-review state rather than claiming exactly once. Remote ticket closure is not verified infrastructure recovery. Permission loss pauses synchronization.

Source review: oshal-kernel/connectors contains all four declarations. ServiceNow currently declares GET resources only; GitHub declares confirmed create/comment actions and issue webhooks; GitLab/Jira resource declarations do not establish complete guarded bidirectional status sync. The actual native write signature takes a confirmed context in crates/connectors/src/actions.rs. Provider completeness, webhook verification and installed admission must be tested per operation, not inferred from ADR prose.

Alternatives rejected: app-owned credentials, blind POST retries, duplicate local ticket engines, last-write-wins status overwrite, feedback loops and automatic incident recovery from remote closure.

Acceptance: One canonical link, correct caller/account and target, denied write, expired/revoked connection, lost response after create, duplicate/reordered webhook, conflict, status mapping, echo suppression and comment redaction. Use disposable provider targets; never create real tickets as part of mockup review or local repository tests.

The first manual implementation is documented in [ADR0025](0025-manual-workflow-ticket-handoff.md). It reserves a current canonical snapshot before confirmed native Actions and labels durable outcomes as operator reports. It does not complete the broader synchronization acceptance above.

ADR 0004: CLI and background work retain user-bound OSHAL authority

Status: Accepted design, 20 October 2026. Provider/runner acceptance remains required.

Context: Kubernetes/cloud/host discovery and evidence collection require tool sessions. A global CLI cache or an administrator's shell default would let one person's credential silently become everyone’s. The experience is based on OSHAL user login and the standard connection framework.

Decision: Resolve admitted connections under the signed-in caller, with explicit named account and resource scope. Interactive connection authorization belongs to the shared secure flow. Unattended work requires explicit activation under an admitted principal and scoped connection; browser tokens and personal terminal caches are not background service credentials. This uses shared authority, not an

independent Operations login. Personal connections and explicitly granted shared connections retain their native ownership distinctions.

Verify runner/tool availability, actual principal, account/tenant/project/cluster incarnation and capability-specific reads. A login does not prove inventory, evidence or write access. Changed binding, expiry, revocation or wrong scope blocks new work. Shared refresh must preserve the bound identity. Tools receive isolated configuration and explicit target selection, without inheriting shell defaults.

Alternatives rejected: application-owned OAuth/token storage, importing global CLI contexts, trusting client-supplied user IDs, all-powerful background credentials and granting writes after a read probe.

Consequences: Connection and delegation metadata appear in topology freshness and collection receipts. Source/webhook authentication still applies independently to background deliveries. The scanner reconciles inventory only after a complete authorized scan; failure cannot delete unseen assets.

Acceptance: Cross-user and cross-account refusal, missing tool, expired/revoked session, withdrawn grants, explicit target binding and no default fallback; prove an observer cannot mutate infrastructure. Use installed shared services and real disposable provider acceptance separately from local UX fixtures.

ADR 0005: Deploy the first persistent experience in an isolated native Kubernetes lab

Status: Implemented first increment, 20 October 2026. Extends ADRs 0001, 0002 and 0004 without superseding their target architecture.

Context: The experience must be tested with actual OSHAL sessions, grants, persistence and Kubernetes observations. The existing browser prototype cannot establish those properties. The full source-principal, detection and automation milestones remain larger than this first increment.

Decision: Package the committed Operations application with the existing pinned native OSHAL runtime. Deploy the application and PostgreSQL in an owned local Kubernetes namespace with persistent volumes, native process sandboxing and loopback access. Preserve the shared kernel and existing incident specialists. Use native reader/operator roles and the kernel's person-scoped application store. No separate login, credential broker or ticket engine is introduced.

Authenticate source registration and imports as the current operator. Save immutable mapping revisions and original/normalized alert records with explicit validation outcomes. Save bounded timestamped pod/container captures, preserving unknown readiness and native UIDs. Expose current-caller connection status through the standard framework. The same caller's records must survive pod replacement; another person must not inherit them.

Alternatives rejected: static mockups presented as a deployment; a second identity service; app-owned cloud credentials; unsandboxed execution to bypass nested Kubernetes prerequisites; automatic ingestion or remediation claimed from a manual import test.

Consequences: This is authenticated capture intake, not a background source identity contract. Exact payload retries share one record, but changed payloads, concurrent decision reservation, retained delivery attempts and the established noise algorithm remain M2/M4 work. Captures are historical

namespace observations, not continuous discovery or a completed correlation graph. Provider/tool activation remains gated by ADR 0004. The initial runtime used a noop model and did not certify RCA inference. ADR0006 records subsequent model-account activation.

Acceptance: Real native login and experience launch; unassigned/read-only refusal; person isolation; downgrade/revocation; actual source/event/quarantine and inventory forms; desktop and phone layouts; saved records after pod replacement; hash-verified real lab firing/resolved deliveries correlated to exact observed pod UIDs. Deployment and acceptance receipts identify the tested source revision. Production users, grants and default Kubernetes context are unchanged.

Owner boundary: Application UI, records, contracts and isolated deployment harness belong here. Any missing connector, delegated runner or transactional primitive belongs to its shared owner and must be verified before dependent milestones are marked complete.

Installation follow-through, 20 October: the kernel-owned local installer now starts a fresh kernel and database, sets the existing administrator's first password once, and loads selected public applications afterwards through the native loader. Initial credentials, desktop profile and application roles remain native framework concerns. The actual OSHAL Node portal window passed password sign-in with an isolated Kubernetes profile; existing bare-metal configuration was preserved. Token import is a separate unfinished shared-runtime contract: the three desktop vendor import routes returned 404 under an authenticated administrator. No token transfer or automatic desktop-worker enrollment is claimed. See IMPLEMENTATION-STATUS.md and kernel `deploy/k8s/local-install-evidence.json` for revision-specific checks.

Subsequent delivery: ADR0006 supersedes the unfinished Codex/Claude import observation above with real installed transfer and model verification. Antigravity and infrastructure-worker acceptance are not implied.

ADR 0006: Select independent applications and shared communication transports

Status: Accepted and partially delivered, 20 October 2026.

Context: The hackathon installation needs Operations, Office, email and Telegram alerts, with Jarvis as its assistant. The broad Jarvis experience package requires Finance and many unrelated applications. Initial Kubernetes desktop transfers failed because native import routes were absent, although the bare-metal satellite had worked.

Decision: Install the independent public `intelligent-operations`, `presentations` and `email-summarizer` packages after the kernel. Use built-in Jarvis and Workflow Studio. Trust the captured public origin before previewing and installing each exact artifact. Keep source identity, native sessions, account transfer, application grants, connections, notification preferences and delivery in the kernel's shared frameworks.

Operations retains its operator role; Jarvis gets chat-user. Office and email currently use application tiers. Optional model accounts are imported only through the signed-in installation operator; a device enrollment credential cannot authorize the import. Choose the model explicitly after testing it. Shared runtime configuration is not permission to impersonate another user's mailbox or to perform an

infrastructure action.

Telegram is a separate native notification transport over fenced HTTPS. It follows the person's saved topic, numeric destination, mutes and quiet hours. A manual test send needs native confirmation and delivery is recorded only after Telegram acknowledges a message. Reuse the existing bot for outbound configuration without changing its webhook. Mail application credentials and mailbox sign-in are separate steps; the connector installer does not enable Google portal auto-provisioning.

Alternatives rejected: installing the entire umbrella bundle; application-private tokens; database writes that bypass registry trust or role previews; treating a stored credential, configured bot or HTTP success as proof that an external message was delivered.

Acceptance: Five active packages with no Finance; admitted app views; real desktop Codex and Claude transfers and model responses; actual locally generated Office artifact; visible disconnected-mailbox guidance; Telegram caller isolation, confirmation, bounded payload and acknowledgement tests. Keep live delivery, mailbox connection, automatic event notification production and full platform publication explicitly separate.

Owner boundary: The store owns Operations' event producer and UX. The kernel owns the installer, notification rail and credentials. M2/M4 still own ingestion, filtering and the established reduction algorithm; this installation does not complete those milestones.

21 October implementation refinement: the shared native notifier now has source-tested SMS and explicitly enabled short phone alerts (core ADR186/188). Brief calls use fixed escaped speech followed by hangup, a30-second ceiling and no recording; interactive Calling Assistant remains separate. Configure services through the standard private Kubernetes connector installer, and save destinations under the original person's topic preferences. The operator authorized exactly one SMS, one brief call and one Telegram test to existing destinations. The operator-side acceptance runner fsyncs each reservation before any possible send, holds lost acknowledgements and mutes test topics. Live acceptance remains pending the final kernel gate/rollout. No automatic alert campaign is authorized by that test.

ADR 0007: Durable processing before enrichment; anomaly detection is separate

Status: Accepted design, 20 October 2026; services pending implementation. Extends ADR0002 after review of the archived processing source and operational notes.

Decision: Preserve the established order: authenticated named source → durable raw delivery and separate telemetry → versioned mapping/identity → stored-query and window evaluation → scoped symptom reduction/reservation → bounded temporal correlation → conditional evidence packet → canonical ticket workflow → standard connector delivery and reconciliation. Repeated occurrences update counts before any new capture or model invocation is admitted.

A separate anomaly service learns from bounded telemetry histories. Its output is an alert candidate containing detector identity/version, resource identity, window, baseline, score, confidence and readiness. That candidate enters the same filtering/reduction path as a source alert. Cold, stale, missing or failed detectors are explicit states, never normal health. Retain externally supplied anomaly results without

claiming that importing or rescoring them provides local detection.

Use native OSHAL identity, current-caller authorization, connection grants and canonical tickets. Application-owned services must not invent parallel users or bypass shared connector operations. Do not launch arbitrary functions or a separate function platform for inbound adapters. Keep raw deliveries, normalized alerts, telemetry, decisions, reservations/outbox and evidence references as distinct durable records, with retention and per-source quotas.

Atomic admission/reservation must survive replicas and restart. A process-local cache is an optimization only. A delivery with uncertain external outcome stays pending reconciliation rather than being marked failed and blindly retried. Rule/backend failure is distinct from no match. Use compound cursors plus bounded overlap and idempotent writes for late/tied telemetry results.

Correlation uses parameterized queries over scoped immutable asset IDs and time-valid edges. Bound depth, visited nodes/edges, elapsed time and output. Record direction and relationship policy. A failure may use a documented field-grouping fallback, clearly marked degraded.

The evidence collector has narrowly granted read access. The investigation worker receives only the redacted packet and earlier phase outputs, without kubeconfig, source credentials or general cluster tools. Script proposals and human acceptance of the report do not authorize execution. Execution remains a separate registered action with exact artifact/target/current-authority checks.

Alternatives rejected: deduplication only at the ticket sender; inferred source routing; unbounded graph joins; treating threshold alerts as anomaly detection; treating collection failure as healthy; memory-only reservations; silently advancing past failed records; silent automatic resend.

Consequences: the hackathon has explicit failure paths and fewer claims of completeness. Service dependencies and all acceptance cases are tracked in PIPELINE-REVIEW.md and PROJECT-PLAN.md. Backend selection/pinning and detector calibration remain implementation work, not implicit approval of any archived infrastructure or credentials.

ADR 0008: Reuse the reviewed topology structure with temporal identity

Status: Accepted design, 20 October 2026. Bounded implementation and native lab acceptance are recorded in ADR0009 and the topology receipt; scheduled discovery and full P4 remain open. Basis: the operator explicitly directed review of the archived correlation database and load scripts. Extends ADR0002, ADR0004 and ADR0007.

Decision: Use the host/service, AWS network and Kubernetes topology loaders as the initial entity and relationship model. The source trace and normalized relationship directions are recorded in [GRAPH-STRUCTURE.md](../GRAPH-STRUCTURE.md). Retain collection batching, source provenance, full-scan generations and explicit typed relationships. Keep the new application vocabulary neutral.

Use native resource identities scoped to their authorized account/project/cluster. Names and IP addresses are aliases, not identity. Preserve actual Kubernetes owner-reference chains. Maintain observation and validity times for nodes and edges so correlation reads the topology relevant to the event. A completed,

scoped generation may retire missing observations; partial or failed scans may not. Stamp freshness only on records actually observed in that scan.

Separate containment, ownership, placement, routing and evidenced dependency relationships. Correlation traverses an explicit versioned direction/type policy with budgets. Sharing a region, datacenter or broad network hub does not alone establish a common incident.

Alternatives rejected: rebuilding the domain structure without reviewing the existing loaders; copying their estate-specific configuration; using pod names as instance keys; deleting graph relationships before a partial reload; stamping every old record fresh after one scan; treating all historical links as causal dependency or choosing the database engine from a document title.

The application owns the normalized graph contract and correlation policy. The kernel owns identity, scheduler activation, caller-bound connections and authorized execution. Database engine and hardware compatibility remain implementation validation, not implicit approval of an archived image or deployment.

Acceptance: provider/account collision, same-name pod replacement, real owner chains, namespace selector matching, autoscaling and movement, partial/failed scan preservation, interrupted generation replay, stale edges, direction, cycles, shared hubs, event-time topology and traversal limits. These remain M3/P4 acceptance gates and must be evidenced through the installed experience.

ADR 0009: Implement the bounded temporal graph on native scoped SQL

Status: Accepted implementation decision, 20 October 2026. Extends ADR0008.

Use the kernel's existing caller-scoped transactional SQL service for immutable inventory scan records and saved correlation decisions. Reconstruct the selected observation time in application code with explicit directed relationships. This provides the reviewed domain structure without adding an archived graph database image or a separate identity service to the installation.

Each collection belongs to a named source. Provider, native account/project/cluster scope, kind and native resource ID determine an asset identity. A complete scan replaces that collection's current observations; partial scans preserve previously observed facts with their original timestamps. Immutable generations preserve history and permit late delivery. A repeated scan ID with different content or two distinct scans at the same collection timestamp is refused. Retries are transactional.

Correlation joins only affected assets connected by an explicitly allowed directed dependency path. An unaffected common hub does not merge its dependents. Partial collections, missing/stale assets, unavailable relevant relationships and exhausted traversal budgets produce a visible degraded result with exact-asset grouping only. Decisions record the policy, selected deliveries, graph revision, scan IDs and paths. Saving a decision does not dispatch a ticket or authorize execution.

This first implementation admits 256 assets and 512 edges in a 64 KiB scan, reconstructs at most 200 scans / 2,000 assets / 5,000 edges, and bounds traversal depth and visits. Exceeding a bound fails explicitly. It does not silently truncate a graph or certify an incomplete result. Production

retention/checkpointing and a larger graph backend remain open; the common contract isolates that later choice. A collection that exceeds a bound must be narrowed before capture.

The initial loader is an operator-run, guarded Kubernetes lab capture. It reads actual native UIDs, ownerReferences, container runtime IDs and EndpointSlice targets, excluding arbitrary pod fields. It is not a periodic scanner or a worker with general cluster access. AWS/Azure/GCP discovery, source conflict resolution, scheduled collection under service grants, automatic suppression integration and original-event-time policy selection remain subsequent work.

Validation: real PostgreSQL transaction/isolation/browser contracts plus installed native API acceptance. PostgreSQL results alone do not establish native deployment acceptance.

ADR 0010: Publish immutable rules before transactional symptom reduction

Status: Implemented in source; installed acceptance is recorded separately in IMPLEMENTATION-STATUS.md. Extends ADR0002, ADR0007 and ADR0009. Owner: Operations package; P2/P3/M4.

Use an actual OpenSearch percolator behind the existing kernel `engine.exchange` relay. The operator pins the package engine endpoint. The package derives its owner scope from the current verified user, checks the engine build digest, and exposes only health, publish and match commands. The engine has no Kubernetes credentials or command execution interface. General HTTP egress exceptions and another kernel adapter are unnecessary.

Publish 1–128 bounded stored queries into an immutable owner/policy index, validate every write and count, then seal the index before activating its SQL source binding with a revision check. Matching exhausts all result pages with an exact count. Timeouts, partial shards, missing indexes, build mismatch and transport failures are failed attempts; they never mean an unmatched event. An unsuccessful activation can leave an unused immutable index; cleanup/retention remains required.

Within one caller-scoped transaction, record the rule decision, symptom occurrence, current count, processing state and evidence-work reservation. Identity includes policy family and the validated source/environment/resource-kind/scope/native-ID/alert-type tuple. The source tuple remains distinct from graph-based incident membership. Replaying the same delivery under another revision of that policy family does not count it twice or create another reservation. Renaming the policy creates a new family and must be a deliberate operator choice.

Window mode is explicitly fixed or sliding, from 1 to 86400 seconds. New episodes and a rise above an episode's highest severity reserve evidence work before expensive collection or investigation. Event time prevents late and same-time conflicting states from overwriting the latest source state. Receipt time controls the suppression window. A resolved source event can update its existing symptom even when firing-only rules do not match; it does not attest recovery or close a ticket. A reservation records whether its matched rules request a snapshot, and never authorizes execution.

Reject copying process-local reservations, partial persistent keys, successful-looking backend fallbacks, first-page-only matches, or suppression after model investigation from the archive. Retain its scoped

key/window/reserve-before-dispatch process with durable transaction semantics.

Acceptance uses real PostgreSQL, two independent SQL pools, real OpenSearch and the shipped browser assets: pagination, disabled rules, rejected publication, revision races, 100 occurrences/one reservation, replay, owner isolation, window boundaries, escalation, late resolution, rollback and lost commit acknowledgement. Named protocol fault seams separately test timeout/partial-page refusal. Installed native SQL and relay behavior require their own Kubernetes receipt.

This increment is operator-triggered evaluation of recorded normalized alerts. Durable OTLP telemetry, time-window aggregation, anomaly detection, background queue consumption, evidence reservation consumers and ticket dispatch remain separate delivery gates.

ADR 0011: Restore the standard shared workspace installation

Status: Standard installation restored (`a3de02d`, `18b68cd`); artifact/approval implementation published (`aac99fb`). Installed experience acceptance is tracked in IMPLEMENTATION-STATUS.md. Owner: native installation and workflow runtime. Extends ADR0001 and ADR0005.

The operator requires the existing OSHAL shared workspace, not an Operations-specific file service. The reference Helm chart already declares `oshal-workspace`, retains it on Helm uninstall, and mounts it at `/app/workspace-shared` for API and bot pods. The native Kubernetes installer omitted that contract. Restore the claim and workspace environment aliases in the standard installation, with a storage-only upgrade that preserves users, applications, provider settings and existing claim capacity.

Kernel and bot workers use that claim. Each worker stages only its admitted task in its private node directory; the agent sandbox does not expose the whole share, node enrollment Secret, provider credentials or Kubernetes credentials. RWO storage requires co-location on its node. Use RWX storage for workers distributed across nodes. The retention annotation does not protect a PVC from namespace deletion.

The workflow produces deliverables through its existing executor. The native broker produces an answer artifact. Named CLI workers already capture actual file changes, a change manifest and the answer with hashes. Do not parse prose into pretend files, substitute assistant-written scripts, or present supplied proposal inputs as generated remediation. Preserve the producing ticket, work, phase, bot and content provenance.

The standard runtime stores accepted output and projects its exact bytes onto the PVC before advancing the workflow. Downloads retain original-caller access checks. The UI must distinguish unavailable storage/results from an empty workspace and must not invent an installed Code Server endpoint. Accepted history remains readable after a source revision update under current ownership, active application/bot and uncancelled authority checks; execution still requires the original exact generation.

Approval goes through the native confirmed ticket approval endpoint and displays its returned state. Approving an investigation/report is not evidence that a remediation ran. Exact remediation approval, constrained execution and measured recovery remain separate acceptance gates.

ADR 0012: Application-defined cockpit navigation

Status: Implemented; installed acceptance recorded in IMPLEMENTATION-STATUS.md.

The Operations ticket link opened the global cockpit without `app=intelligent-operations`. The global framework profile also returned every static application tile without checking whether its provider was installed, active and admitted to the current person.

Use the existing application manifest contract. Operations `ui.static` declares its Overview, Topology, Data pipeline, Sources and Workflow Studio destinations; `ribbon.hideFrameworkItems` retains the canonical Tickets view and excludes unrelated framework workbenches. Ticket links preserve `/cockpit/?app=intelligent-operations&ticket=…`. The shell resolves the declared default tool name to its registered view identifier. Workflow Studio appears only when its owning application is active and admitted.

Global toolbar links use the same current route-owner admission as application tiles. An administrator does not get links to unloaded applications merely by being an administrator. Native Files, Security Center, Test Lab and Eval Wall remain kernel pages with their existing reader/operator checks. The Apps menu continues to use current admitted workspace discovery. Hidden navigation never substitutes for route authorization, which is checked again on use.

This changes navigation and context retention; it does not implement missing telemetry, anomaly detection, generated remediation or automated infrastructure execution.

ADR 0013: Inspectable graph and processing cohorts

Status: Implemented and verified in the local Kubernetes experience; receipt in `deploy/evidence/detection.json`.

The experience must explain what survived filtering, deduplication and correlation. Counters without shared provenance or raw JSON alone do not provide that explanation.

The Graph explorer loads the caller's reconstructed inventory and presents searchable, typed, keyboard-accessible nodes. Selecting a node opens its directed relationships, health observation, native identity, scope and source evidence. The overview renders at most 100 matching nodes and states the truncation; source reconstruction budgets remain unchanged. Missing nodes are not inferred to be deleted after partial collection.

The processing flow uses one latest returned decision per delivery. Its incoming cohort, percolator matches and new/escalated symptoms share that denominator. It displays both fractions and counts, with event drill-downs and separate unmatched/suppressed states. These are latest returned pages, not installation totals or a live throughput claim. Policy replay does not increase a delivery count. Correlation coverage is shown separately because analyses can overlap and cannot safely be added together.

An explicit user action evaluates selected events using their active source policy, then correlates only the admitted/escalated survivors. Each existing service retains its own durable transaction; failed evaluations remain visible and retryable. No automatic collector or infrastructure action follows. Saved correlation analyses come from the existing owner-scoped decision journal, survive reload, and show groups, input events, paths, graph revision, time and limitations. Degraded exact-asset fallback is labeled as such, never as proven dependency correlation.

The graph and flow reuse existing source, policy, SQL, OpenSearch, original-caller and native engine contracts. They introduce no separate database, ingestion runtime or credential store. Continuous processing, separate OTLP observations, anomaly detection, conditional evidence consumers, execution and external synchronization remain open.

Native acceptance also exercises shared framework dependencies: cockpit tool URLs must place query parameters before application fragments, and SQL mutations must load foreign-key dependencies of earlier transaction reads before validating the whole snapshot. Neither fix weakens caller admission or referential integrity. The pinned local OpenSearch image is verified by content ID before Kind import aliases are repaired on each node.

ADR 0014: Native scheduled processing of admitted alerts

Status: Implemented; native scheduled processing and pause/resume verified in the local Kubernetes installation.

Recorded alerts must continue through published rules without an operator repeatedly pressing Evaluate. Use the existing OSHAL personal scheduler and declared service route. The activating person remains the caller; each run rechecks current package authority. There is no application daemon, borrowed administrator credential or cluster access.

Processing is disabled initially. The operator selects exact saved source revisions, each with a published policy, saves the configuration and activates the native schedule. Configuration uses revision checks. Pausing the native schedule stops future ticks; disabling processing also fences already claimed work before a decision commits.

The existing transactional queue owns eligibility, attempt count and a 120-second lease. Each bounded run claims one item at a time, processes at most ten, and stops starting items after 45 seconds. A native 120-second invocation budget accommodates the final engine request. Claims, decisions and lease checks retain caller isolation. Concurrent workers cannot count an occurrence twice. An expired claim can be recovered; a stale worker cannot commit a decision. Transient failures receive exponential backoff (up to one hour), permanent or exhausted failures require an explicit retry. A lost commit acknowledgement cannot turn completed work back into pending work.

The experience displays saved processing configuration separately from native schedule activation and recent job states. Missing scheduler access is unavailable, never active. Revoked access clears private records. Native run failures remain failures even if an earlier item in that batch succeeded. Lease tokens are not exposed to the experience.

This increment consumes already admitted alerts and reserves downstream evidence work. It does not authenticate an external producer, ingest OTLP, correlate on a timer, collect evidence, generate

remediation, authorize execution or synchronize tickets. Those remain separate delivery gates in the full Event-to-Action plan.

Acceptance: eight real SQL/OpenSearch/browser contracts cover concurrent workers, expired claims, stopping processing, retry exhaustion and explicit recovery, lost commit acknowledgements, owner isolation and unavailable scheduler UI. The complete Operations suite passes 60 cases with no skips. The installed native scheduler processed seven labeled acceptance alerts without a manual processing call or trigger. A 65-second pause retained new work; reactivation processed it. Both native receipts report scheduled execution and HTTP 200. Acceptance leaves the worker disabled and its schedule inactive. The configuration row is locked through the decision commit so disabling cannot acknowledge ahead of that commit.

ADR 0015: The workspace editor belongs to the kernel baseline

Status: Implemented in the native Kubernetes installer; acceptance is recorded separately.

Decision and reason

Code Server is installed with the kernel, before loading public applications. This restores the existing OSHAL Helm baseline instead of making each application install its own editor. Workflow Studio generates and accepts artifacts; the kernel projects those exact bytes to the standard shared `oshal-workspace` PVC. The editor mounts that same claim and the portal resolves the admitted ticket workspace.

The installer waits for administrator bootstrap, kernel readiness and editor readiness before recording a successful baseline. Existing installations have an explicit editor upgrade that preserves accounts, applications, workers and workspace data. Normal forwarding manages both local browser ports and reconnects after pod replacement.

Access boundary

This is the existing local operator workspace model. The editor has read/write access to the entire share, with no kernel credentials, provider-account volumes or Kubernetes service-account token. The listener binds to pod loopback; the operator reaches it through loopback Kubernetes port-forward. Exec probes check the loopback listener. An ingress-deny policy adds protection where supported, but the local kind CNI does not enforce NetworkPolicy; the loopback listener is essential. Public or multi-user remote editor sessions require a separate authenticated, scoped gateway and are not delivered by this decision. Portal artifact downloads continue to enforce current caller access.

Acceptance

Verify a fresh baseline with no application loading, administrator password login, removed bootstrap password Secret, ready kernel/editor and bound workspace PVC. Upgrade the existing installation and open a workflow-generated report through its portal link; compare its bytes in the kernel and editor containers. Verify that the loopback health endpoint works and direct pod-IP editor access fails. Source regressions pin ordering, isolation, port validation and preservation during upgrade.

Kernel commands and receipts: `deploy/k8s/LOCAL-INSTALL.md` and `deploy/k8s/editor-evidence.json`. This does not establish generated remediation, approved infrastructure execution or full platform certification.

ADR 0016: TriageNorth is the product; the platform is a dependency

Status: Product/foundation boundary directed by the operator, 20 October 2026. TriageNorth is the proposed product name; submission documents updated.

This supersedes the product naming and submission framing of ADR0001, while retaining its shared-framework architecture and stable `intelligent-operations` identifiers.

TriageNorth is the working customer-facing name for a secure AI analysis and response hub. Event-to-Action is the initial domain workflow. OSHAL is the existing platform foundation providing security, workflows, UX, connections and tickets, installed as a versioned Docker/OCI baseline before the product package. Disclose inherited platform capabilities rather than claiming them as hackathon inventions.

The submission emphasizes operational interpretation, explainable filtering/reduction, temporal correlation, bounded evidence, the investigation experience and controlled response. Generic means reusable across supported evidence sources and analysis workflows; it does not mean unrestricted agents or automatically supported connectors.

The commercial positioning centers on application value and optional deployment/support services. Foundation commercial arrangements remain outside this application decision. MARKET-RESEARCH.md documents competitive overlap, public prices and limited cost examples from primary sources; it records hypotheses and differentiates current behavior from predictive and execution roadmap gates.

Acceptance: product-led brief/proposal, explicit foundation boundary in editable Visio, research sources and assumptions, updated plan and exports. Stable runtime IDs and historical acceptance receipts are preserved; changing submission language does not claim a deployed UI rename or new functional capabilities.

Foundation attribution follows the existing project licensing reference: AGPL-3.0-or-later with a paid commercial alternative. Native-kernel metadata conflicts with that reference and remains a baseline publication check, not a new application license decision. See MARKET-RESEARCH.md for source provenance.

ADR 0017: Resource acceptance precedes local pipeline restart

Status: Original stop recorded 20 October; supervised restart authorized 21 October. A necessary workspace-floor guard is implemented in kernel source `d37720e`; 49 focused fixture checks pass. The

resumed local installation passes placement/reserve checks with explicit node budgets. A complete production resource profile remains open. The operator identified separate training as the source of the earlier load.

Context

The operator stopped Docker after repeated memory failures. The retained journals confirm host OOM and NVIDIA allocation errors in two boots. Source and pre-shutdown inspection establish a separate installation problem: the local workspace PVC co-locates services whose permitted memory peaks exceed their kind node's ceiling. The account sidecar must be included in that group. See RESOURCE-REVIEW.md for the observations, exact sources and limits of the causal evidence.

Decision

Keep runtime work stopped and use no GPU workload for this application. Retain the standard shared workspace, kernel authority and isolated investigation model. Before runtime acceptance resumes, require an explicit installation resource profile that fits the host and each node, including local-volume placement, sidecars, system headroom and startup/rollout overlap. Refuse an over-budget plan before starting workloads. Source review and the subsequently authorized small CPU-only unit checks do not authorize restarting Docker.

Do not silently increase limits, enable swap, change host VM settings, disable security controls or launch auxiliary engines to get a passing demonstration. Resource values remain unchanged until a reviewed correction is implemented and the operator permits runtime work. No numerical replacement budget is selected by this ADR. Source edits and acceptance results must remain distinguishable.

Consequences and verification

The full pipeline goal remains open. Historical successful tests remain useful evidence but do not prove current availability or capacity. New source review must not rerun service-dependent hooks, renderers, browser or GPU checks while the stop constraint applies. Publication and generated exports requiring those processes remain explicitly pending; hooks are not bypassed.

Acceptance requires a pre-start refusal for insufficient host/node capacity, no GPU device admission in the application profile, measured startup and steady-state memory, unchanged shared-PVC behavior, no observed OOM/restart regression and the complete event-to-action user journey. Infrastructure execution still requires its own exact-artifact/target approval and verified outcome.

Source increment and remaining boundary

Kernel `d37720e` adds `deploy/k8s/install_capacity.py` and calls it before fresh installation and editor, worker, account-sidecar or workspace resource changes. It compares the combined RWO workspace deployment limits with Docker caps and Kubernetes allocatable memory, respecting node selectors and bound PV affinity. Every possible unbound-claim placement must fit. RWX retains independent pod floors. It counts sidecars, initialization peaks, replicas and declared pod overhead; it never changes limits. All 19 new regression cases pass, with fixture subprocess execution prohibited. The exported committed installer snapshot passes 49 tests across capacity, editor, workspace, account and installer suites. CI registration is present; the full service-dependent gate remains unrun since shutdown.

The fixture scope enforced 25% of one core, 256 MiB RAM, zero swap, 16 tasks, nice level 15 and a 60-second timeout. Actual cgroup values were checked before execution. Peak scope memory was 33.6 MiB; the Python run used 25% CPU over 2.79 seconds. No runtime resource values were changed or services started.

This catches a necessary condition that failed in the observed installation. It does not certify host reserve, other pods, non-container/GPU demand, concurrent changes, rollout overlap, aggregate RWX capacity or measured peak usage. The reported result explicitly carries `fullCapacityCertified: false`. No installer or runtime execution is authorized by committing the source.

ADR 0018: Replay evaluation and bounded local text analysis

Status: Evaluation design recorded following the operator's three-week improvement request. Dataset preparation, a measured BGL percolation baseline and native synthetic replay are implemented. NAB anomaly evaluation and model selection remain pending. Extends ADR0007, ADR0010 and ADR0017.

Decision

Use versioned, hash-identified datasets and frozen replay manifests to evaluate percolation, anomaly detection and reduction/correlation separately. Preserve the original inputs and record transformations. Keep labels outside detector/model inputs and calibration separate from untouched evaluation data. Historical external metrics do not supply or validate the local Kubernetes topology.

Retain actual OpenSearch percolation behind the existing engine relay. Benchmark statistical anomaly baselines separately, then feed detector candidates into the normal alert/filter/reduction path. Thresholds, rules and labels must not be silently adjusted to improve holdout results. Partial backend failures remain failed evaluations, never no-match results.

Evaluate a deterministic summary against small local model candidates for offline mapping assistance and post-reduction evidence summarization. Inbound processing keeps its deterministic identity/format contract and durable reductions before expensive investigation. Model output cannot rewrite authoritative measurements, identity, severity, decisions or labels. It has no execution authority. All runtime model use goes through existing admitted framework provider/worker facilities.

Use existing source/provenance and temporal graph projections for a replay desk, stage counts and an isometric SVG estate view. The latter is a visual-depth design, not a true 3D implementation claim. Keep accessible 2D navigation and explicit observed/inferred/unknown and historical/live distinctions. No GPU rendering or inference is authorized by this decision.

Rationale and alternatives

Dataset replay makes incorrect filtering, missed incidents and false alarms measurable before a polished presentation hides them. Lightweight statistical baselines establish whether additional model complexity earns its resource cost. Reject an LLM call for every raw event, label-assisted evaluation, destructive rewriting of original logs, fabricated topology and local-cost claims without memory, latency, retry and review measurements.

Acceptance and consequences

THREE-WEEK-PLAN.md owns the ordered work, scorecards, dataset sources, candidate models and proposed bounded inference profile. evaluation-data.json records the prepared upstream pilot and explicitly records no detector/percolator/model run. Normal Store checks remain required before publication. Binary exports and the PowerPoint deck remain separately tracked deliverables; adding this ADR does not make them current or complete. Full runtime acceptance remains subject to ADR0017.

ADR 0019: Systems over the current user's observed inventory

Status: First source increment implemented and tested on 21 October 2026. Publication, deployment and browser acceptance are pending. Extends ADR0004, ADR0009 and ADR0013; runtime work remains subject to ADR0017.

Decision

Build the requested Kubernetes/systems inspection experience as a projection of the existing user-scoped temporal graph and event page. Reuse the admitted session, topology and record APIs. Preserve the existing estate toolbar identity and point its Systems label at the new view; detailed topology remains linked inside it.

Filter by provider/scope, namespace, resource kind and observed health. Retain native identity and capture time on every resource. Use recorded graph edges for placement and ownership, and exact unambiguous resource identity for related events. Keep unknown health, partial collection and bounded results visible. An identity match is not a causal diagnosis or evidence of recovery.

Reason and consequences

The operator wants useful inspection without duplicating data already entering the application. A second collector, local credential store or privileged shell would add a new boundary without improving this first inventory journey. Those are excluded from this increment; future scoped diagnostic operations must use the standard connection and tool framework.

The existing graph and latest event page limit coverage. Source freshness and missing observations remain visible; this view does not imply continuous collection or live cloud discovery. Data and authorization failures clear the new view rather than leaving apparently current records on screen.

Acceptance

Source `1e7141f1` implements the view and twelve focused behavioral tests, including the actual controller's refusal and late-response handling. See [Systems](../SYSTEMS.md) for the implemented journey, limits and pending browser/server checks. Measured investigation time and collection overhead belong in the pilot; no commercial savings percentage follows from adding the view.

ADR 0020: Three rolling context banks and durable evidence

Status: Accepted native profile, bounded growth and real-time minimum-age/packet-survival canary.
Date: 21 October 2026.

The operator recalls a three-database rotation strategy that preserved extended data while alerts passed through filtering. Alerts alone omit the context an investigation needs; wholesale rotation can erase it while a ticket is still processing.

Use three logical, caller-scoped telemetry banks in the standard application store, with explicit resource mappings and configurable byte/record bounds. Rotate on capacity; never reuse a bank whose newest accepted receipt is younger than 900 seconds. Transactional backpressure is preferable to acknowledging a batch that silently destroys protected context. A 3×5-minute schedule alone fails the minimum-age requirement.

Keep telemetry, the alert ledger and pinned evidence separate. A durable matched capture reservation authorizes a bounded context projection for its exact resource. Copy that projection and provenance into an immutable packet before releasing the bank lock. Packet retention is independent of raw rotation and does not confer execution authority. The first native profile uses three SQL collections rather than three database servers; it reuses platform scope and atomic checked commits instead of introducing a privileged database-management plane. Bounded safe bank growth and reviewed ceiling increases preserve existing rows. The byte/record ceiling remains a backpressure boundary. Physical storage autoscaling requires measured profiles and remains open.

Accept a bounded OTLP JSON log/numeric metric profile and decoded CloudWatch subscription batches through existing caller authorization. Require native ID/scope mappings and CloudWatch account/group/stream bindings. Use standard collectors to receive syslog/file logs and decode provider transport. Never infer a machine identity from a display name.

Provide separately labeled recorded research samples and deterministic synthetic scenario controls. Exercise actual percolation, reduction and graph paths; retain raw provenance, licensing notices and original timestamp ties. A synthetic cohort is a regression and demo, not evidence of detector accuracy on real datacenters.

Required verification: concurrent writers/retries, changed event ID conflict, whole-batch rollback, 900-second reuse boundary, forced owner policies, exact resource windows, pinned evidence surviving rotation, actual engine replay, browser refusal clearing and native migration/role admission. Source rollback uses a new revert commit. Existing telemetry/PVC data is retained; reverting source must not drop its tables or packets.

Native verification on 21 October accepted actual 909.788-second reuse, stable pinned packet hashes after reclamation, reviewed bank growth and a pinned standard Collector export. Saved replay cohorts reuse the existing owner-scoped application store: only immutable decision/correlation references are accepted, and the server reconstructs counts and the exact graph revision. Reopening a cohort does not replay deliveries or confer new capture/execution authority. This is a durable demo/audit view, not a new unattended workflow or a live-health assertion.

ADR 0021: Versioned rule editing and side-effect-free candidate trials

Status: Implemented in source; focused real SQL/OpenSearch/browser acceptance passes. Date: 21 October 2026. Extends ADR0010 and ADR0013.

An operator needs to understand and change the percolation wall without rewriting a JSON policy from scratch or changing live incident counts merely to test an idea. The application keeps a named rule form, enable/capture choices, explicit fixed/sliding suppression window, simple AND-condition builder and lossless advanced query JSON. Saved immutable generations can be loaded as drafts; activation is always explicit.

A candidate trial selects 1–20 validated deliveries from the same owned source. Reject foreign-owner/source and invalid references before contacting the engine or saving a candidate. Seal a complete query generation through the existing real percolator and save it as an intentional immutable candidate in the existing policy journal. Return complete match sets per delivery; no active binding, filtering decision, symptom count or evidence reservation changes. Repeating the same definition reuses its generation. These saved candidates are retained policy history, not disposable unmanaged indexes.

Activation separately compares the source binding revision that the editor loaded. A changed binding preserves the draft and blocks a stale overwrite. Existing decisions retain their original policy; reevaluation is a separate recorded action. Trial mode never grants snapshot or execution permission. The same current-user records-write authority covers saving candidates and activation; the browser clears private rule, history and trial contents on access withdrawal and ignores late results.

The focused acceptance uses actual PostgreSQL owner policies and OpenSearch, not a local query imitation. Browser acceptance exercises editing, trial counts, explicit activation, saved-draft restoration, phone layout and denial clearing. This decision closes the rule-adjustment interaction; it does not itself wire continuous monitoring, generate remediation, execute a change or establish incident recovery.

ADR0022: 3D exploration over the authorized temporal graph

Status: source/browser tested and native24d62980 accepted; see deploy/evidence/operating-experience.json. Date: 21 October 2026.

The product needs a usable 3D graph with predefined operational questions. Render the existing current-user, selected-time graph in a CPU canvas with deterministic XYZ layout, perspective depth, orbit, pan, zoom, directed edges and node selection. There is no continuous animation, external CDN or new data authority. Keep the existing 2D typed-column view and equivalent keyboard-accessible node list.

Presets: observed estate; outgoing dependency/routing paths; incoming dependents (potential impact); bidirectional placement/ownership; explicitly degraded observations; observations older than 15 minutes relative to the graph's selected time. Dependency traversal is at most four hops and 500 visited identities;

placement is two hops. Display is at most 100 nodes and 500 edges, with truncation visible. Unresolved targets are not fabricated as resolved nodes. Unknown health is distinct from healthy; historical observations do not become live probes.

Existing topology GET admission and database owner isolation remain authoritative. The browser runs predefined traversals over that already bounded response; these are snapshot presets, not arbitrary server-side database queries. Refusal, refresh and access withdrawal dispose observers and remove private pixels and node lists. No inference, cluster access or infrastructure mutation is authorized by exploring a graph.

Proof: graph-space.test.cjs covers known directed graph answers, cycles, stale time, unknown health, invalid identities, bounds, deterministic depth and real shipped desktop/phone interaction. Existing SQL/temporal graph and Systems tests continue to exercise ownership and late-response clearing. Native acceptance is recorded independently in deployment evidence.

ADR0023: Link evidence to the canonical investigation workflow

Implemented and native24d62980 accepted, 21 October 2026. The operating-experience receipt proves actual first-phase inference and shared-workspace output.

An evaluated, admitted symptom with a capture reservation can pin its preprocessed context and prepare a canonical Intelligent Operations ticket through the original user's native ticket service. The application stores one durable association per immutable reservation. It records a creating marker before the external create call, deduplicates concurrent requests and holds unknown outcomes for reconciliation rather than blindly creating another ticket. A lost link acknowledgement reconciles the already saved exact ticket ID.

The operator starts investigation from Data & decisions. Preparation returns one fixed evidence/packet.json, canonical JSON bytes and SHA-256 digest, plus the bounded original-owner dispatch input. The browser uses the existing canonical ticket start API with the user's portal session. Existing workflow phases and shared workspace persistence generate the accepted answer artifacts. No separate incident engine, cluster credential, raw stream handle or infrastructure executor is introduced.

Funnel metrics join exact delivery/reservation/ticket IDs. An AI assessment counts only a completed first-phase native work record with a released nonempty answer and inference receipt. Canonical done means report acceptance. Verified infrastructure recovery remains an explicit unconnected stage until a real recovery-check receipt exists. Unknown/unreadable work remains unavailable, not zero; unrelated tickets and repeated evaluations cannot inflate the cohort.

The UI refreshes workflow progress on request and clears private content on access refusal. This increment does not add automatic background investigation dispatch, multi-file remediation generation, execution approvals, recovery verification, external ticket status sync or outbound messaging. Those are distinct active build gates, not implied by a created or completed ticket.

Proof: real PostgreSQL tests cover concurrent preparation, owner isolation, unknown creation and lost commit acknowledgement. The actual shipped browser verifies sealed-byte dispatch, idempotent resume, receipt-based counts and refusal clearing against an explicit canonical-service fixture. Live native inference and workspace acceptance are recorded separately after deployment.

ADR0024: Service standing is distinct from account and delivery status

Accepted, 21 October 2026; source and native password-login acceptance pass.

Connections must show what the signed-in user can actually invoke. A local credential or saved connection is insufficient evidence of a configured native executor or a successful message. The Operations view reads standard GET /connectors and /api/notify/prefs. It shows selected endpoints, original-person invocation standing, usable/expiring personal accounts, registered transports and enabled personal routes. It never displays SMS/Telegram destinations or credentials. Legacy kernel metadata remains an explicit rollout requirement rather than a false connected state.

The central HTTP gap is documented in kernel ADR187, and central SMS in ADR186. No app-specific credentials or transport are added. The view performs no effect. Phone readiness remains separate from SMS; provider health remains not-probed. Refresh generations discard stale results, and any authorization refusal clears all private service rows. Two actual browser cases prove this behavior.

ADR0003 still owns durable external-ticket direction, phase/status mapping, operation reservation, reconciliation and echo/conflict behavior. This standing view does not claim those domain contracts are implemented.

The development installation now registers SMS, brief phone and Telegram, and selects Jira under core ADR187/188. Separately authorized one-time tests have transport acceptance; they are not inferred from this read-only screen. Jira's profile read returns HTTP 404, so live account validation remains open. ADR0026 records these acceptance boundaries and the outstanding full kernel gate.

ADR0025: Manual workflow step to external issue handoff

Accepted,21 October2026; source SQL/browser checks pass, installed acceptance pending.

Use immutable original-person links to existing Jira, GitLab or ServiceNow records. The link binds provider identity, the exact selected personal account, endpoint/auth revision and explicit status:phase mappings. Canonical ticket type, status, phase and revision come from the existing owner-checked ticket port. The application stores no credential or caller-selectable URL, header or operation template. At most32 links are admitted per owner under a serialized SQL capacity guard.

A person reviews the mapped operation, then confirms its push. The application reserves that exact snapshot plan before dispatch. The browser rechecks current service realm/account and canonical revision, uses the standard fresh native confirmation and connector Actions, and reads back the external state. An already matching state makes no external write. Lost responses remain held; read-only reconciliation never authorizes another transition for the same plan. A workflow can advance during a distributed request; this is a manual snapshot handoff, not an atomic cross-system transaction or an automatic production subscription.

Native Actions provides credential/owner/tier/marketplace admission and protected endpoint binding. The browser's readback is shown immediately, while durable application outcomes remain explicitly operator-reported and verifiedByServer=false. A forged client report cannot establish native delivery, mutate canonical status or inflate recovery metrics. Server-attested receipts and unattended reconciliation are separate work. Declared one-way mappings do not imply webhook verification, bidirectional changes, comments, conflict resolution or echo suppression.

The UI includes Jira transition discovery, phase/status mapping controls, saved-link review, held attempts and read-only reconciliation. GitLab uses numeric project and issue IDs with close/reopen; ServiceNow uses an explicit state value and its expected displayed value. No default state numbers are guessed. Splunk On-Call needs reviewed credential handling; GitHub status handoff is not added here.

Proof: actual SQL forced-owner isolation, concurrent reservations, lost commit acknowledgement, bounded link capacity and stale revision rejection. Actual shipped browser tests exercise native-API seams for review/confirm/readback, lost provider response without duplicate write and withdrawal during final refresh. These are isolated fixtures; no external issue was changed. Core ADR187 separately proves fixed provider transport over actual owned TLS through native Actions.

Named account selection is passed as the standard Actions account query and enforced by the kernel selection ladder, including its connection tier and original owner. A kernel without advertised named-account support is refused before dispatch; a UI preflight alone is not account binding.

ADR0026: Record local service acceptance separately from release certification

Accepted, 21 October 2026.

The operator authorized one SMS, one brief phone call and one Telegram test to the existing private destinations. These are original-person tests through the shared native notifier. They do not enable a general automatic alert campaign or authorize external issue mutations.

The owned local Kubernetes installation now runs development kernel `ac5f00e`. The build uses an immutable source export and the existing verified runtime base; image/binary digests and a private rollback snapshot are retained. Only the kernel image and source annotations change. Accounts, sidecars, environment, placement and volumes are compared before/after. Supported service keys enter through the standard private connector installer. Credentials and destinations never enter the product package, public receipt or agent evidence.

The acceptance runner fsyncs an attempt reservation before any possible send and never resends an attempted channel. SMS and voice returned explicit carrier acceptance. Telegram's successful HTTP response used the installed older contract; its newer-field checker initially marked it unconfirmed. The record was reconciled against that exact legacy success contract, explicitly marked as an inference, without another send. All three channels have one attempt and all test topics are muted. The operator subsequently confirmed receiving the phone call in chat. That confirmation is recorded separately from the native carrier acknowledgement; The operator reported no SMS. A read-only provider lookup uniquely matched the existing test and returned `undelivered`, error `30034`: the sending number is not associated with an approved US A2P 10DLC campaign. No additional message was sent. SMS

registration/delivery remains blocked; Telegram human receipt is unverified. The provider definition is documented at https://www.twilio.com/docs/api/errors/30034.

The installed connection/notification screen and existing three workflow-generated shared-PVC reports pass native password-login checks after the development update. Jira is selected and connected under the original person, with its invocation scope and named-account selection. Its read-only profile probe returned HTTP 404; live account/site validation remains open and no issue mutation is claimed.

Full release certification remains open. The wider committed kernel gate found a stale SQL cache (fixed centrally in core ADR189), stale fixture contracts/counts (corrected without waived assertions), and a 264 MiB stream that expired its ten-minute handle under the test load. The complete rerun on `af5afcb` remains pending. The development image is explicitly marked uncertified; the kernel PR remains draft. This local acceptance does not turn a failed or incomplete full gate green.

Sanitized evidence lives in `deploy/evidence/communications-acceptance.json`, `development-kernel.json`, `services-standing.json` and `jira-readonly.json`.

ADR0027: Make the evidence-to-decision handoff inspectable

Accepted, 21 October 2026; installed UI acceptance passed. Fresh report acceptance is recorded separately below.

The application seals a bounded preprocessed packet and dispatches the canonical queue workflow. The operator could see report files but not the exact inputs, inter-phase additions, named bot handoffs or useful graphical progress. Core ADR190 supplies that reusable visibility through the ticket's existing work port, without expanding agent access to the raw streams or Kubernetes.

The specialist output contract now requires an Evidence reviewed table citing actual input paths and JSON fields, a Decision table with supporting facts and stop conditions, and bounded diagnosis/verification or explicitly approval-gated remediation proposals when the evidence supports them. The reviewer checks those claims against the submitted packet. The submitted prompt requests the same structure. The output allowance is6144tokens within existing native ceilings; accepted output remains the workflow-generated answer.md, not invented extra files.

The current canonical ticket view must show the exact admitted packet hash/content, new or changed files per phase, actual bot/receipt/tokens and current graph position. Reference-cost estimates are labelled separately from recorded spend and display the installation's configurable assumption. Past reports are not rewritten as if new instructions had applied to them. A fresh bounded acceptance run must prove new report quality separately.

Create is the broader public studio experience, with AI Office/presentations as its document/deck component. Install it through the trusted native loader and use current user authorization. Preserve existing Office files and do not install optional image/video/model runtimes merely to expose the studio home.

Intelligent Communication mailbox notices link to the Email/mailboxes catalog, which keeps actual Google and Microsoft sign-in separate from administrator SMTP or webhook transport for portal email alerts. Microsoft client configuration can reuse the existing reviewed private installer; mailbox consent and a matching provider callback remain explicit acceptance steps. No email send is authorized by this UI/configuration change.

Installed development acceptance at kernel 6f1c38f passed actual HTML basic login, positive separately labelled reference cost, exact original-owned inputs, named bot receipts, the ticket graph and read-only Studio graph on desktop/phone, Create home and the filtered Gmail/Outlook catalog. The UI baseline used genuine existing three-phase ticket ca7fc196; its past outputs are unchanged. Create, Operations and Communication were installed from committed application bceffaf0 through native repository trust/preview/consent and user grants.

A new bounded monitored incident admitted 8 actual observations from an explicitly synthetic local workload, with captured pod UID, timestamps and hashes. Its native ticket 2a60f877 failed at 45 seconds; the receiver's private status confirms `codex_timeout`. This failure is preserved. The central account deadline correction at 782ae3d permits90 seconds under the105 second HTTP/120 second phase ceiling. The original packet is reused only through an explicit linked child; no alert replay, raw access or infrastructure execution is authorized. A missing successful usage receipt cannot establish zero provider consumption.

Microsoft authorization URL generation is verified using the reviewed private installer input. Personal mailbox consent, provider callback registration and an administrator SMTP/webhook transport remain unverified; no email was sent.

Fresh installed acceptance now passes: linked child 537d9d04 completed all three analysis phases and awaits human report approval. The original packet and both earlier reports reach revise. All three shared-workspace reports passed native authenticated download/size/hash checks.10 browser/API checks pass; see the sanitized `deploy/evidence/decision-chain.json` receipt. Generated scripts remain proposals inside the reports; standalone script delivery and approved execution are separate unfinished gates.

ADR0028: Bind declared conversational bots explicitly

Accepted and installed, 21 October 2026.

A weekly RCA request submitted through the actual Jarvis UI reached the Operations specialist, then failed with `operation has no explicit permission binding`. The kernel admitted the application and completed Jarvis inference, but refused the specialist before its inference. No new workflow was created. HTTP-only acceptance had missed this separate non-HTTP admission boundary.

Bind both declared Operations bots by name to the existing `app.open` read permission. Its resource scope remains `own`; existing roles, write permission, workflow approval and execution authority are unchanged. The regression enumerates every declared bot and requires its explicit binding and the own-scope read role. Do not bypass the kernel refusal, add a broad role or fabricate a successful job.

After normal publication, use the existing trusted native package install and composite-role reapproval flow, then submit through Jarvis again only after checking for any partial work. The report, PDF, PowerPoint and email are separate results and require actual artifact and provider receipts.

Source `2ef275de` was published and merged in PR547. The exact installed Operations artifact is `dc3d158459d6a92cbbac3e6c9b2bc301a307e22f9efde11a49171bb236670e56`. The real UX retry completed specialist and Office inference. Subsequent operator evidence export supplied the own-account records; this is not autonomous evidence retrieval. Create's separate studio installation and the actual PDF/PowerPoint/mail receipts are recorded in ADR0029 and `deploy/evidence/create-suite.json`.

ADR0029: Install Create's declared studio components

Accepted, 21 October 2026. Native export and provider acceptance verified.

Create 1.15.0 was already installed and matched repository main. Only Image Editor and AI Office were available beneath it: the optional studio packages had not been installed. A complete experience requires both package admission and the declared Create composite role with the selected optional applications.

Use the shared guarded operator helper `scripts/install-create-k8s.py` to install Video Studio 1.11.0, its required Vids 1.5.7 package, Publishing Studio 0.1.0 and PDF Workbench 0.1.0. Keep the existing Office, Create and Operations packages. Use exact public committed source, trusted native preview/install, and Create's own administrator template with only those optional components selected. The Publishing/PDF engines use existing reviewed local images, verified source hashes, bounded CPU/memory and the native private relay. No GPU workload is started.

Actual Video upload finalization initially failed: the Kubernetes scratch volume had mode `02777` and its newly created private root inherited mode `02700`. The editor requires exactly `0700`. An owner-checked initialization step clears inherited SGID on that one private directory before kernel startup; no permission check is weakened. A real host fixture reproduces inheritance and refuses a symlink. Live pod replacement preserves the corrected `0700` root. The actual Create UX imported two one-second clips, saved a revision and exported 60 frames at 1280x720/30fps. Independent decoded samples verify the blue/orange splice.

Jarvis generated an evidence-based Word report after an operator supplied a bounded authenticated export of eight own-account ticket records. The operator then passed that saved outline to the actual native Publishing and Office APIs: Publishing generated an eight-page PDF with clean preflight; Office generated eleven PowerPoint slides including the cover. Bytes, signatures, retained facts and hashes were independently checked. These are system-hosted application artifacts, not manually manufactured local report files. This acceptance does not establish a fully automatic Jarvis evidence-to-multiple-artifact chain or placement in a canonical ticket workspace; those integrations remain open.

After the operator registered the callback and personally completed Google consent, Gmail accepted two messages containing the requested PDF and PowerPoint, one attachment per message through the existing native mail API. Provider IDs and the explicit recipient remain in private operator receipts. This proves provider acceptance, not human inbox receipt. No SMS, call or Telegram was resent.

Kernel replacement temporarily interrupted the portal-facing Code Server redirect. The editor deployment stayed ready. The existing forwarding service reconnected; the kernel/editor deployments and editor service endpoint are ready and the editor health endpoint returns HTTP 200. No kernel source change was needed for this slice.

ADR0030: Plan Git-backed Jarvis delegation and promotion

Status: proposed roadmap, 21 October 2026. No hierarchy implementation is claimed.

The operator requested Git connectivity and upper/lower swarm cooperation: repair and prove changes in a lower environment before promoting them to the upper one. Adopt the staged scope in [the promotion roadmap](../SWARM-PROMOTION-ROADMAP.md). Use standard user-bound connections, canonical linked tickets/workspaces and registered swarm identities. Preserve distinct identities and authority at every hop. Promotion binds approval and evidence to an exact commit and build digest.

Reusable delegation, authority, durable transport and recovery belong in the platform where gaps are demonstrated. Incident policy and the visible promotion journey belong in TriageNorth. Existing ticket synchronization is not Git change execution. A child's successful result is evidence for an approval, not a grant.

H1–H5 define delivery dependencies and positive/refusal acceptance. This decision adds planning and connection shortcuts only; it enables no autonomous deployment.

ADR0031: Separate mitigation, permanent repair and recovery verification

Status: proposed design, 21 October 2026. Extends ADR0030's roadmap.

Support four operator-selected response paths: paired mitigation/permanent repair, standard lower-environment promotion, bounded emergency runtime patch, and an expedited Git-backed production hotfix with explicit omitted-check waivers. Keep separate linked change records under the incident. A runtime patch requires Git reconciliation; an expedited Git deployment uses an exact reviewed artifact.

Start post-change watching as a durable, bounded, read-only workflow branch using versioned health criteria. Command success does not resolve an incident. Healthy, regressed, inconclusive and timed-out outcomes are distinct. Conditional rollback requires exact prior authorization plus current preconditions, or a new approval. A temporary fix's deadline prompts review rather than blind reversion.

[Response design](../RESPONSE-PATH-DESIGN.md) describes screens and implementation acceptance. The interactive prototype makes no live calls and grants no authority.

ADR0032: Give operators one resource workbench and one navigation owner

Installed and browser-accepted, 21 October 2026. Native navigation, resource ownership expansion and historical RCA citations are visible in the portal.

The native cockpit supplied an application rail while the embedded Operations page rendered its own header and full navigation. The Systems view used repeated cards instead of a persistent selection/detail layout, and its default screen used promotional wording. These are concrete sources of inconsistent navigation.

When embedded, the native rail now owns navigation; the inner header/nav are hidden. Standalone access retains its own shell. Native entries cover Systems, topology, data, live context, scenarios, sources, connections and credentials. Systems is the default; Overview remains an explicit summary. Shared theme tokens and compact page headings apply across Operations pages. No authority follows from a navigation entry, and no private/unloaded application is added to the toolbar.

Systems uses a bounded exact-identity tree and one inspection panel. Grouping rows are explicitly groups. Workload ownership chains determine primary containment; node placement remains an observed relationship. Ambiguous ownership and cycles are not silently resolved. Capture time, unknown health and partial collections remain visible. Read-only graph refresh does not pretend to probe a cluster. The lab collector additionally retains measured pod phase/readiness and container waiting/termination facts without copying environment credentials.

The AI terminal and historical RCA reference library are separate backlog work. Native generic HTTP services currently enforce a public HTTPS fence and expose no Kubernetes private-CA/exec adapter. Do not weaken that fence or grant cluster credentials to the evidence-analysis bot. A reusable original-user infrastructure connection/execution service is required. Existing RCA outputs remain canonical ticket work/artifacts; the current Operations manifest declares no RCA reference collection. A future library must cite and revalidate those sources rather than invent a second causal graph from report text.

ADR0033: Project historical RCAs from their canonical sources

Installed and browser-accepted, 21 October 2026: catalogue, original work receipts, input lineage and accepted report links are visible.

Add a historical RCA library over current-user canonical tickets, completed work receipts, input lineage and accepted workspace artifacts. Preserve one source of truth for reports. Reuse the Systems split-panel layout and native navigation. Do not claim the topology correlation tables are a reviewed RCA case library.

Keep report acceptance separate from infrastructure recovery and preserve partial lineage or unavailable sources. Require exact original-ticket download selectors, render untrusted report text safely, and clear private views on refusal or stale selection. The default Systems refresh also uses the same resolved page key as navigation so an empty/unknown hash cannot erase its only graph read.

[RCA-LIBRARY.md](../RCA-LIBRARY.md) records current limits and subsequent scoped retrieval work. There is no automatic RAG ingestion, cross-case cause inference, terminal execution or expanded bot authority in this increment.

ADR0034: Inspect pods through the shared original-user connector

Implemented, 21 October 2026. Native status/command execution and namespace RBAC refusals pass on the owned cluster; installed browser inspection and exact command review/execution also pass. Runtime receipts identify checked sources.

Use Kubernetes API exec for pods. Keep cluster selection, certificate validation, credential vaults, namespace guards and execution admission in the shared kernel. The Operations application provides a consistent resource-detail console and exact command review. AI interprets selected normalized results through the existing accounted bot rail and receives no cluster credential or execution capability.

Use bounded commands initially. Require fresh confirmation, identify the observed pod UID and selected container, invalidate review on edits, and represent uncertain transport outcomes as unknown. Preserve native Kubernetes RBAC alongside the installation namespace allowlist. Never treat transport closure as proof of remote termination or report acceptance as recovery.

[POD-CONSOLE.md](../POD-CONSOLE.md) describes current behavior and remaining persistent terminal and historical retrieval scope. Shared framework rationale and configuration belong in kernel ADR191 rather than duplicated application services.

ADR0035 — Capture retained context on the personal processing schedule

Status: implemented in source; native deployment acceptance pending.

The evaluator committed evidence reservations but required a separate manual pin before the three raw banks rotated. The existing personal processing schedule now offers an explicit **Preserve retained evidence after filtering** setting. Existing configurations default to off. This authorizes only retention of already admitted data for the selected source revisions.

A bounded consumer follows evaluation and also drains earlier pending reservations. Each completion locks its processing item and current configuration, validates its source and reservation, then commits the immutable packet, reservation outcome and processing outcome together. No duplicate capture occurs across concurrent consumers. A rule without capture records `not-requested`; absent context records `no-retained-context`, never invented observations. The packet preserves its source, time, content hashes, redaction and bounded truncation indicators.

Failures use durable exponential backoff and the configured attempt ceiling. Explicit retry resets only capture; it does not repeat filtering or increment suppression counts. Disabling the worker serializes with an in-flight capture commit and prevents later ones. Schedule pause controls future ticks; disable controls the active commit fence.

This consumer uses the existing native personal schedule, original-user SQL scope and application permissions. It grants no infrastructure access, model call, external communication or execution authority. Automatic correlation, collection from external systems and canonical workflow dispatch retain separate acceptance gates. The current native package ticket port exposes create/list/get and needs a shared start operation for unattended workflow dispatch.

Validation: real disposable PostgreSQL consumer tests cover opt-in, ownership, concurrent completion, no-capture rules, backoff/exhaustion/retry, pause serialization and redacted retained context. The existing real OpenSearch processing test covers repeated occurrences through reduction into one immediately eligible packet. Native schedule acceptance must be recorded separately before claiming installation completion.

ADR0036 — Consistent daily operator screens

Status: accepted direction; implementation and deployment receipts are recorded separately. Date: 21 October 2026.

Problem

Operations, Systems, Data pipeline and Sources mixed explanation, daily monitoring, setup forms and raw records. Pipeline exposed the rule editor, worker settings, correlation and every event in one long page. Sources rendered every revision as a full card followed by both setup forms. Internal hash navigation could leave the cockpit rail naming another screen.

Decision

Use a shared compact operator layout over existing caller-owned records. Operations highlights attention and recent activity. Systems keeps its tree and selected resource detail. Pipeline has Flow, Events, Rules, Automation and Correlation tabs. Sources has search, a selectable revision list and its exact mapping/detail panel; setup is a separate disclosure.

Cross-screen actions use the existing manifest-owned `app-navigate` message with bounded source/event selectors. The shell remains responsible for resolving an authorized application screen. Standalone navigation uses local query parameters. Do not introduce a second navigation service, credential store, collector or action executor.

Accuracy and authority

Counts are explicitly bounded to returned cohorts. Source registration does not imply connectivity. Captured health is not a live probe; report acceptance is not recovery. A tab switch does not trigger evaluation, publication, collection, model calls or infrastructure actions. Original action controllers remain responsible for writes. New read projections clear on refusal and reject stale async ticket responses.

Verification

Browser contracts cover source search and exact-revision drill-downs, event trails, raw-data disclosure, tab keyboard behavior, phone widths and access withdrawal. Existing SQL/browser contracts continue to exercise policy trials and activation, personal worker configuration and sealed investigation start. Native installed-screen inspection is a separate acceptance step.

Embedded Systems follow-up

Native cockpit inspection at 1280px exposed inherited setup-form label margins and an early filter wrap that displaced the resource workbench. The filters now use compact margins and a single desktop row at embedded widths of at least 960px, with wrapping below that width. Filtered tree groups use the validated system catalogue for their labels, preserving the observed cluster name without adding excluded assets. Tests exercise the 1040px embedded width and filtered-pod label alongside existing phone, keyboard and access cases.

ADR0037: Approved response, isolated execution and verified recovery

Status: implemented in source, 22 October 2026; SQL and shipped-browser guards pass; installed acceptance on the owned lab is recorded separately in IMPLEMENTATION-STATUS.md. Extends ADR0011, ADR0023, ADR0025, ADR0031 and ADR0034.

The response stage closes the gap between an accepted report and a measured recovery without adding cluster authority to any bot or package service.

Generated files are extracted byte-exact, never invented. The workflow's accepted answer (the revise phase's actual `answer.md` work artifact) names its response files as fenced blocks whose info string is `file=NAME`: `remediate.sh`, `verify.sh`, `rollback.md`, `verify.md` (optional `diagnose.sh`). The package stores those bytes with SHA-256 digests, byte ranges, the producing ticket, work record, phase and answer digest. The answer bytes reach the package through the operator's browser from the canonical work read, so the record is labeled `answerProvenance: operator-forwarded-canonical-work`; the kernel has no package port for work reads or artifact registration. Prose is not parsed into files; a missing file, a duplicate, an oversize script or a `remediate.sh` without a preceding "approval required" statement makes the answer non-executable, and the UI says so.

A plan binds one observed target. The target pod is accepted only when the caller's own recorded inventory (the temporal graph) has observed its UID and name; namespace and container are explicit. The plan identity is the digest of ticket, canonical revision, work record, answer digest, artifacts and target. The exec input is fixed at plan time: `/bin/sh -eu -c <remediate.sh>` inside that container, which keeps the executor within the namespace-scoped pod exec RBAC the standard pod-connect installer grants.

Approval binds principal, digests, target, revision and expiry. An approval is recorded by the signed-in person against the plan's artifact digests, target and canonical ticket revision and expires after thirty minutes. A changed revision or digest refuses approval and dispatch before anything runs. Report acceptance remains a separate canonical gate (ADR0011).

One dispatch per approval, reserved before the native action. Reservation precedes the browser's fresh native confirmation and `kubernetes-exec` through the person's own Kubernetes connection (ADR0034). Twelve concurrent callers reserve once; a lost commit acknowledgement leaves the reservation in place and refuses a second dispatch. The pre-dispatch identity readback must return the same pod UID. The action result the browser reports is stored as an operator report (`verifiedByServer: false`); it never changes canonical status or counts as recovery.

Recovery is measured, with four outcomes (ADR0031). A verification is a reserved read-only exec of `verify.sh` plus a current pod readback. `regressed` when the check fails, `inconclusive` when its outcome is unknown, the identity differs or no resolved signal exists yet, `timed-out` after fifteen minutes without one, and `healthy` only when the check passes, the identity matches and a validated `resolved` delivery for the same resource (and alert type, when the plan is linked to an investigation) was admitted after the action. The signal half is read by the server from the caller's own deliveries. Up to ten verifications per execution.

The ticket update is server-read. After a healthy outcome the browser approves the canonical report gate with a fresh confirmation; the package then re-reads the ticket through its ticket port and records status and revision as `verifiedByServer: true`. External issue handoff stays the manual snapshot flow of ADR0025.

Downloads. Every artifact and the execution receipt are authenticated own-scope downloads with `Content-Disposition: attachment` and the digest in a response header. They are package records, not kernel ticket-workspace artifacts: the kernel materializes only worker results.

Alternatives rejected: parsing prose or supplied lab proposals into files; a package-held cluster credential or kubectl executor; approval without expiry or digest binding; counting command exit as recovery; automatic re-dispatch after an unknown outcome; changing canonical status from a client report.

Proof: `tests/execution.test.cjs` (real PostgreSQL forced-owner isolation, concurrent and lost-acknowledgement reservations, expiry and revision invalidation, outcome rules with firing versus resolved deliveries, server-read ticket update, and the shipped panel against explicit native confirmation/exec/approve fixtures).

ADR0038: Local anomaly service, automatic correlation and tick notifications

Status: implemented in source, 22 October 2026; SQL and detector guards pass; NAB pilot receipt recorded in `deploy/evidence/anomaly-nab.json`. Extends ADR0007, ADR0014, ADR0018 and ADR0020.

A real detector, local and explicit. Two deterministic statistical detectors ship in the package (`ui/anomaly-detector.js`, shared by the server tick and the offline evaluation): a rolling robust z-score (median and MAD over the last N samples) and a seasonal time-of-day profile (per-bucket median and MAD). Each new metric sample is scored against the baseline learned from earlier samples only, then learned from unless it was anomalous. Readiness is explicit per resource and metric: cold (warm-up), ready, stale (gap larger than the configured cadence budget, which restarts the rolling baseline), late (older than the cursor, counted and skipped), invalid, failed. No imported or rescored external result is counted as detection.

Candidates are source alerts. A candidate is admitted through the normal intake under a dedicated alert source revision the operator chooses: `alertType = anomaly:<detector>:<metric>`, status firing or resolved, severity by score, and the raw payload carries detector id/version, window, baseline, score, confidence, readiness and the telemetry observation id. Candidates then pass the same percolation rules and symptom reduction as any source alert, so a published policy for the candidate source decides what becomes work.

Durable, idempotent processing. Detector state lives in `operations_anomaly_state`: the configuration (compare-and-set), one compound cursor per telemetry source `(observed_at, id)` locked for update per pass, and bounded baselines (at most 500 per person, at most 2000 samples per rolling window, 64 per seasonal bucket). A pass reads at most 500 samples per source in cursor order, admits candidates (idempotent by candidate id), and advances the cursor only after every admit of the batch succeeded; a failure keeps the cursor, records the error on it, and the next tick retries. Two concurrent replicas cannot both admit the same candidate.

Automatic post-reduction correlation. When the processing worker opts in, each scheduled tick that admitted or escalated a symptom correlates the admitted firing cohort of the policy window over the recorded temporal graph with a bounded versioned dependency policy. The cohort's newest alert time keys the record, so an unchanged cohort is idempotent; events without a recorded asset stay explicitly unresolved; nothing is dispatched.

Tick notifications. After each tick the package asks the kernel notification rail (`uses: notify.send`) to tell the signed-in person about admitted symptoms, candidates and failures. The rail chooses the channel from the person's preferences and may skip silently, so the tick reports a request, never delivery.

Measured, not tuned. `lab/nab_evaluate.cjs` evaluates both detectors on the pinned NAB pilot: a-priori parameters on whole series, and parameters calibrated on the first 30% of each series with labels unread, evaluated on the remainder after a one-lookback gap. The receipt reports detected and missed windows, first-detection lead time, alerts outside windows per observed day, warm-up and runtime, and says it is not an official NAB score. The a-priori run detects every labelled window with zero false alerts on both controls for the seasonal profile, but about twenty alerts a day on the noisy CPU series; the calibrated run brings that to below one a day at the cost of the EC2 latency windows and a much longer lead time. Both are recorded; neither is hidden.

Alternatives rejected: threshold rules presented as anomaly detection; importing archived detector results; a cursor by timestamp only; learning from anomalous samples; silently advancing past a failed admit; counting a notification request as delivery.

ADR0039: Node-level Kubernetes explorer and coached commands under the operator's own authority

Status: implemented in source, 22 October 2026; guards pass. Extends ADR0019, ADR0032 and ADR0034.

Nodes are first-class in Systems. The scan normalizer records each node's capacity and allocatable (CPU, memory, pods), its conditions with any pressure flagged, cordon state, roles, taints, internal IP and hostname, kubelet, OS, container runtime, architecture and kernel, as bounded scalar facts. The detail pane shows a capacity table, the pressure and cordon warnings, the version facts and the pods placed on the node from recorded placement edges; each pod is a link that expands the tree to the exact pod. Absent facts are omitted, never fabricated, and the pane says the facts are saved observations, not a live read.

Coached commands run under the person's own authorization. The Command helper page asks the operations analyst for one reviewable command from a `cli-helper-v1` packet: tool, goal, optional context (treated as data) and the recorded target. The reply is an argv array with reason, read-only flag, risk and verification; prose without a valid argv stages nothing, and a proposal that looks like it contains a credential is discarded. Two execution paths exist, and the page says which applies:

• `pod-exec`: the reviewed command runs once through the person's own Kubernetes connection with

a fresh native confirmation (ADR0034), and the native result is recorded.

• `kubectl`, `aws`, `gcloud`, `az`: the person copies the command and runs it in their own

terminal or OSHAL Node under their signed-in profile, then records the pasted result. The record is labelled operator-provided; the application executed nothing and holds no cloud credential.

Records are closed, bounded (16 KiB output, 200 per person) and own-scope.

What the kernel would need for in-application cloud execution. There is no rail today for a page or package to run `aws`, `gcloud`, `az` or `kubectl` under the person's credentials: the cloud connector catalog has no executable AWS signing (SigV4), no Azure management connector, no SSH or host-shell connector, and the native node cannot receive shell tasks. Those are listed in [KERNEL-RAILS-NEEDED.md](../KERNEL-RAILS-NEEDED.md); this application will adopt a rail the day it exists without changing its review-first flow.

Alternatives rejected: a package-held kubeconfig or cloud profile; running commands through the analysis bot; presenting pasted output as verified; hiding nodes behind pod placement only.

ADR0040: Git promotion and break-glass reconciliation

Status: implemented in source, 22 October 2026. Package guards pass; the kernel catalog operations are on `dev/native-parity` as 0f86a3c with their own guard. Live acceptance needs an installation that ships that catalog, the GitHub or GitLab service selected by an operator and the person's own connected account. Extends ADR0025, ADR0031 and ADR0037.

Roger asked for two levels of change: an upper level where changes go to Git and a pipeline applies them, and a lower level where an emergency patch is pushed directly (break-glass). Both now start from the same response plan (ADR0037), so the bytes that are reviewed, executed and committed are identical.

Standard path (upper level). From a plan, "Commit, open request" reserves one promotion per plan and destination, then the browser, under the person's own GitHub or GitLab connection with a fresh confirmation per write, reads the base branch, commits the plan's exact generated files plus a manifest (plan, ticket and revision, work record, answer digest, target, artifact digests) to a new branch

`triagenorth-<plan>`, opens a pull or merge request whose body states that nothing was executed on the cluster, and optionally starts the pipeline on that branch (GitHub workflow dispatch or a GitLab pipeline). Merge and pipeline status are read back on request.

Break-glass path (lower level). The emergency change itself is the existing approved exec (ADR0037). After it has a reported outcome, "Reconcile break-glass change in Git" commits the same files with the execution receipt to `triagenorth-reconcile-<plan>` and opens a request that must be merged, or reverted in Git, within 24 hours of the execution; the deadline is recorded and an unmerged reconciliation past it is shown overdue. A reconciliation cannot be reserved before an execution exists.

Authority and evidence. The server holds the file bytes and builds the manifest, commit message and request text; the browser only relays them through the kernel's connector actions, which fix the provider origin, admit exact variables and require confirmation for every write (kernel 0f86a3c). One reservation per plan, path and destination is written before the first provider write; a retry is refused rather than creating a second branch, and a failure mid-way is held as uncertain. Provider results (commit, request number and link, pipeline status, merged) are the operator session's report (`verifiedByServer: false`); links are accepted only on github.com or gitlab.com.

Swarm-to-swarm. When a lower swarm owns the monitored environment, the same manifest and files are the natural delegation payload; routing them over Jarvis between installations needs the kernel A2A rail listed in KERNEL-RAILS-NEEDED.md and is not claimed here.